

4. Calculate the Effective Nuclear Charge for the following: (3 marks each - 6 marks total)

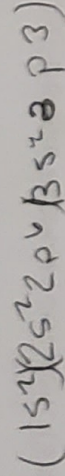
For an s or p orbital, sum:

- 0.35 for each electron in the same group
- 0.85 for each electron one shell lower
- 1.00 for each electron 2+ shells lower

For a d or f orbital, sum:

- 0.35 for each electron in the same group
- 1.00 for each electron in any other lower group

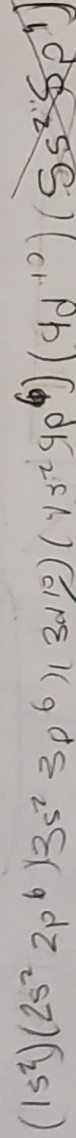
a) a 3p electron in phosphorus (3 marks)



$0(1 \times 2) + (0.85)(8) + (0.35)(4) = 5 = 10.2$

$Z_{\text{eff}} = 2 - 5 = 13 - 10.2 = 2.8$
 $\therefore Z_{\text{eff}}$ is 2.8 for 3p electron

b) a 4d electron in tellurium (3 marks)



$36 - (1) + (0.35) \cdot 9 = 36.35$

$36 - 36 + 0.35 \cdot 9 = 3.15$

$Z_{\text{eff}} = 36.35 - 33.15 = 3.2$

PART C: APPLICATION (12 marks)

Answer the following questions in the space provided.

1. Label each of the following sets of quantum numbers as *allowed* or *not allowed*. Identify the problem for each of the not allowed sets. (3 marks)

a) $n=3, l=2, m_l=0$

Is allowed

b) $n=5, l=1, m_l=3$

Not allowed since

$-l \leq m_l \leq l$ No $m_l=3$

lower exist for p orbital

c) $n=1, l=2, m_l=-2$

Does not exist because l=0 orbital does not exist, minimum is 3d.

2. Identify the element whose highest energy electron would have the following four quantum numbers: (2 marks)

3, 2, +2, +1/2

Manganese

6, 1, +1, +1/2

Bismuth

3. Write the four quantum numbers which describe the last electron in each of the following: (2 marks)

Bromine $(4, 1, 0, \frac{1}{2})$ ✓

Zirconium $(4, 2, -1, \frac{1}{2})$ ✓

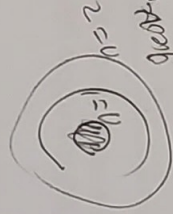
0 5
3 3 3 3

4. An element's position on the periodic table can indicate its chemical properties. This relates to the arrangement of electrons within an atom of that element. Answer the following questions, referencing the concepts of electron configuration and/or energy level diagrams, and Z_{eff} .

- a) Which element has the larger atomic radius, chlorine or bromine? Explain your answer. (2 marks)

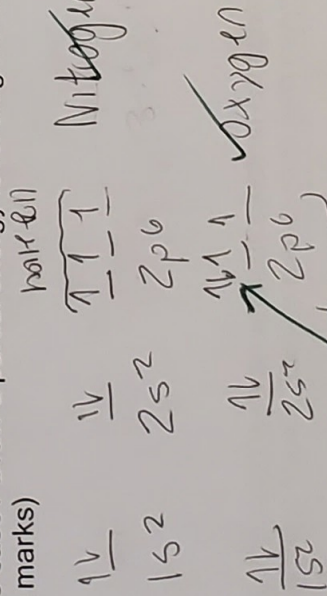
Bromine has the larger atomic radius as even though they Bromine has more protons, its shell is one level away leading to a larger atomic radius. Going down a period results in larger atomic radii. Looking at

Z_{eff} is ↓
experiences more shielding!



valence shells and n values for quantum numbers
3p5 vs. 4p5
4 > 3, higher valence, further away.

- b) The first ionization energy for nitrogen is 1402 kJ/mol, and for oxygen it is 1314 kJ/mol. Why is this the case? Draw a partial energy level diagram of the valence p subshell to aid in your explanation. (3 marks)



Neither half is full

repulsion is stronger for 2nd e^- in pair so easier to remove.

Nitrogen In this scenario, nitrogen has a half-full 2p subshell meaning it wants to keep it that way for lower energy. Oxygen on the other hand has a lone electron in the 2p subshell making it easier to remove, but the values are still close because oxygen gained a proton which increases ionization energy. That is

why Nitrogen has a greater ionization to as to remove from a half-full shell requires more energy than removing from a shell that is neither half-full (111) or full (111111).

2.5

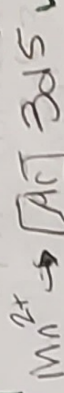
Answer the following questions in the space provided.

- $$\Delta E = \frac{h \nu}{\lambda} = \frac{(6.626 \times 10^{-34} \text{ J} \cdot \text{s}) (\checkmark 3 \times 10^8 \text{ m/s})}{500 \times 10^{-9} \text{ m}}$$

$$2975 \times 10^{11} \text{ J}$$

$$(w_{e_direct})$$

- $$\frac{10}{25} = \frac{2}{5}$$



✓
[Ne] 3s²3p⁶
— 1W
4+

- [illegible]

the expected configuration

Athens' principles

- observed/actual/anomalous

configuration -

However, ~~cont~~
 ② 13 is more stable
 as it fills the 4d
 subshell first hence
 Since filling 4d first
 has a lower potential
 energy state. That is why
 ② is the actual
 configuration.

bulky